Harshit Vaibhav Barla

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.77
NARAYANA HIGH SCHOOL	Class XII (CENTRAL BOARD OF SECONDARY EDUCATION)	2023	94.40%
NARAYANA ENGLISH MEDIUM SCHOOL	Class X (CENTRAL BOARD OF SECONDARY EDUCATION)	2021	76.80%

EXPERIENCE

Software Engineer Intern

360 Technologies, Scadea

- Developed inspection modules for the **Kavalan AI** enterprise security gateway, enforcing **Data Loss Prevention (DLP)** policies and blocking unauthorized AI-service access to sensitive corporate data.
- Built **REST APIs** using **Rust** and **Axum** for real time telemetry and **policy management**, enabling reliable integration between security services and frontend dashboards.
- Implemented **mitmproxy based plugins** for authorized **TLS traffic inspection** and automated payload filtering, **improving visibility** and **policy enforcement** across monitored network requests.

Software Development Intern

Soul AI, Deccan AI

- Evaluated Gemini and other LLM outputs across **1,000+ prompt-based test scenarios** using structured rubrics, identifying recurring failures in reasoning, instruction-following, and response consistency.
- Automated batch evaluation and regression-testing workflows using **Python and SQL**, comparing model performance across task categories, detecting dataset inconsistencies, and guiding targeted prompt and data-quality improvements.

Software Engineer Intern

Lampro Tech LLP ,Visakhapatnam, INDIA

- Contributed to an **OCR-based CAPTCHA recognition pipeline** through image preprocessing, dataset preparation, and model evaluation, achieving **96.7% accuracy on a held out test dataset** and executing training jobs through **AWS Batch**.
- Implemented image upload and inference APIs alongside responsive React components, adding **input validation** and **error handling** to improve prediction workflow reliability.

PROJECTS

RouteLoop - Adaptive LLM Gateway & Evaluation Platform (*Github Link*)

Go, TypeScript, Python, PostgreSQL, Redis, ClickHouse

- Built a full-stack multi-provider **LLM gateway** for **OpenAI**, **Anthropic**, and **Gemini**, processing **10K+ requests** with dashboards for tracing, cost/latency analytics, model comparison, routing policies, and failure investigation.
- Developed **evaluation-driven** routing across **4+ workloads**, combining quality, cost, latency, and reliability signals to **reduce benchmark inference cost** by **30%+** with <1% quality loss.
- Implemented **1K+ request replay**, **2% shadow traffic**, and **5 to 25 to 50% canary rollouts**, with **automatic**

rollback and dashboards for experiment performance and cost savings.

FraudLens CRM - AI Powered Sales & Risk Management Intelligence *(Github link)*

Python, React.js, TypeScript, PostgreSQL, Docker, LLM APIs

- Built a **full-stack CRM platform** with real time **deal tracking, lead management**, and **AI-powered insights** that analyze customer interactions and predict deal success probability.
- Implemented **LLM-based intelligence** to summarize sales conversations, extract intent and sentiment, and recommend next actions, enhancing decision-making and lead prioritization.
- Designed a responsive **React dashboard** with **Kanban pipelines**, **analytics charts**, and real-time updates using **WebSockets**, and deployed the system with **Docker** and **CI/CD pipelines** for scalable cloud execution.

SKILLS

Programming & Database: Python, C, C++, Go, Rust, TypeScript, JavaScript, SQL, PostgreSQL, Redis, ClickHouse, DBMS

Frameworks & Libraries: Axum, Pandas, NumPy, Matplotlib, PyTorch, Scikit-learn, WebSockets, Socket.IO, Jest

Cloud & Tools: AWS Batch(AWS (Bedrock, Kinesis, S3, STS, IAM), Docker, CI/CD, REST APIs, mitmproxy, LLM APIs (OpenAI, Anthropic, Gemini)

RELEVANT COURSES

- **Computer Science :** Data Structures, Algorithms, DBMS-1, Compilers-1, Theory of Computation, Operating Systems-1.
- **Machine Learning & AI:** Foundations of Machine Learning, Predictive Modelling, Artificial Intelligence, Multi-Agent Systems (MAS), and Convex Optimization.
- **Pure Mathematics:** Math Foundation, Linear Algebra, Real Analysis, Functional Analysis, Algebra I - Groups and Rings, Introduction to Group Theory, Introduction to Number Theory, Probability Theory, Multivariable Calculus, Calculus-I & II, Ordinary Differential Equations, Partial Differential Equations, Complex Variables, and Numerical Analysis.

POSITIONS OF RESPONSIBILITY

- **Pronites and Production Head**, ***Milan'25*** - **Headed** the Pronites & Production team of **12 coordinators**, overseeing **artist management**, **budget planning**, **production logistics**, and crowd management of **1000+ attendees**.